

MODULE 1

Introduction: Basic communications model - Protocol layers and service models - Basic definitions - OSI model - Internet protocols. Application Layer: Network application architecture, Web, HTTP, FTP, SMTP, POP3, and DNS

A computer networks is a collection of computers and other hardware interconnected by communication channels that allow sharing of resources and information. A network is a group of devices connected to each other.

INTRODUCTION : BASIC COMMUNICATION MODELS

Communication Model:

The basic communication model in computer networking is where the Sender (encodes the message) channel sends a message over a channel or medium and receiver (decodes the message) gives Feedback.

Components:

The components involved in the successful implementation of the basic communication model are as follows –

- Sender – Who sends the message.

- Encodes – Translates messages into symbols like words, pictures, sound, etc.
- Channel or medium – It used to transmit messages. Some channels are face-to-face communication, over telephone, letters, television, newspapers, radio, etc.
- Decode – Receiver decodes these symbols to understand what the sender wants to say.
- Receiver – A person who receives the message.
- Feedback – After receiving a message, the receiver sends feedback to the sender, answer and what he understands from the message.

Sometimes noise is also part of the communication process and it disturbs the message and it will be difficult for the receiver to understand the exact message that the sender wants to send. Communication can be verbal, written or by gestures.

In the basic communication model, the sender encodes the message and transmits the message through communication channels like verbal (face-to-face, over telephone, video calls) and non verbal (newspapers, letters) for sending the message.

Then, the receiver receives the message and decodes it according to their understanding and again sends feedback to the sender.

Communication Model

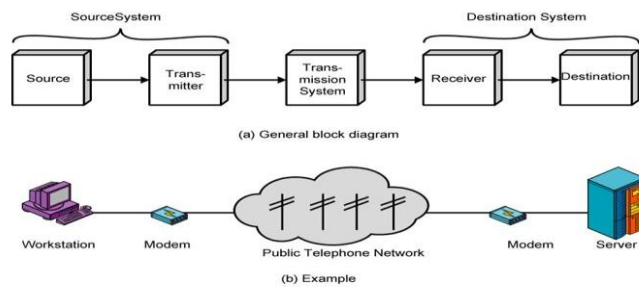


Figure 1.1 Simplified Communications Model

Protocol layers and service models

PROTOCOL LAYERING :Protocol layering is a common technique to simplify networking designs by dividing them into functional layers, and assigning protocols to perform each layer's task.

For example, it is common to separate the functions of data delivery and connection management into separate layers, and therefore separate protocols. Thus, one protocol is designed to perform data delivery, and another protocol performs connection management. The data delivery protocol is fairly simple and knows nothing of connection management. The connection management protocol is also fairly simple, since it doesn't need to concern itself with data delivery.

Protocol layering produces simple protocols, each with a few well-defined tasks. These protocols can then be assembled into a useful whole. Individual protocols can also be removed or replaced as

needed for particular applications.

- Computer networks typically follow a layered architecture, commonly referred to as the OSI (Open Systems Interconnection) model or the TCP/IP model. These models consist of different layers, each serving specific functions. Here's a brief overview:

OSI Model:

1. Physical Layer: Deals with the physical connection between devices.
2. Data Link Layer: Responsible for node-to-node communication, error detection, and framing.
3. Network Layer: Manages routing, addressing, and logical communication between devices.
4. Transport Layer: Ensures end-to-end communication, flow control, and error correction.
5. Session Layer: Establishes, maintains, and terminates sessions or connections.
6. Presentation Layer: Handles data translation, encryption, and compression.
7. Application Layer: Provides network services directly to end-users and applications.

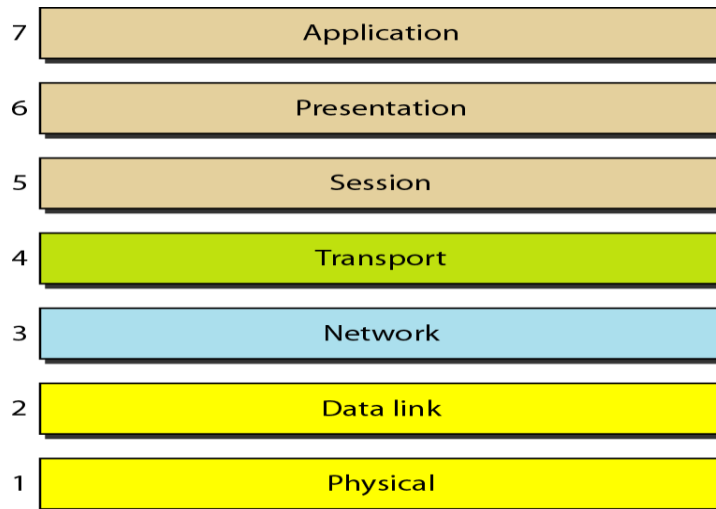
TCP/IP Model:

1. Link Layer (equivalent to OSI's Data Link and Physical layers): Involves hardware addressing, network topology, and error detection.
2. Internet Layer (equivalent to OSI's Network layer): Manages packet routing, addressing, and fragmentation.
3. Transport Layer (similar to OSI's Transport layer): Ensures end-to-end communication and reliable data transfer.
4. Application Layer (combines functionalities of OSI's Presentation and Application layers): Provides network services directly to applications.

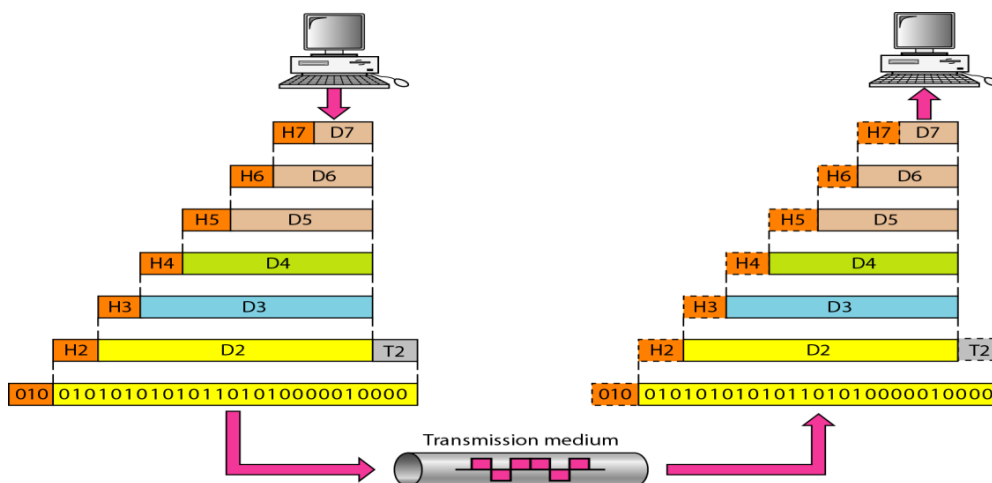
THE OSI MODEL

Established in 1947, the International Standards Organization (ISO) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (OSI) model. It was first introduced in the late 1970s.

Seven layers of the OSI model



An exchange using the OSI model



1. Physical Layer

Coordinates the function required to carry a bit stream over a physical medium. The physical layer is responsible for movements of individual bits from one hop (node) to the next.

The physical layer is also concerned with the following:

- Physical characteristics of the interfaces and the medium : Defines the characteristics of the interface between the devices and the transmission medium.
- Representation of bits : Defines type of encoding of bits (conversion to signals).
- DataRate: Rate at which data transmitted
- Line configuration – It refers to the way two or more communication devices attached to the link
 - * point _to_ point : Provide a dedicated link between two devices.
 - * multi _point : More than two devices share a common link
- Physical Topology : A Network Topology is the arrangement with which computer systems or network devices are connected to each other. Bus, Star, Ring ,mesh etc are commonly used topologies.

- Transmission mode

- 1.Simplex Mode –**

- In Simplex mode, the communication is unidirectional

- Only one of the two devices on a link can transmit, the other can only receive.

- 2.Half-Duplex Mode –**

- In half-duplex mode, each station can both transmit and receive, but not at the same time.

- 3. Full-Duplex Mode –**

- In full-duplex mode, both stations can transmit and receive simultaneously

2.Data link layer

Responsibilities of data link layer

- Framing :Divides stream of bits received from the physical layer into manageable data unit
- Physical addressing :Data link layer adds a header to the frame to define sender/receiver
- Flow control : If the rate at which data are absorbed by the receiver is less than the rate at which data is produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
- Error Control : Data link layer is responsible to detect and retransmit damaged and lost frames and also handling frame duplication.
-
- Access Control : When two or more devices are connected to the same link, data link layer protocols are

necessary to determine which device has control over the link at any given time.

3.Network Layer

Network layer is responsible for source-to-destination delivery of a packet. Responsibilities of the network layer

- Logical addressing : If the packet passes network boundary, network layer add logical address of the sender and receiver.
- Routing : When independent networks are connected to create a large network route packet to the final destination. One of the functions of the network layer is to provide this mechanism

4.Transport Layer

Transport layer is responsible for process to process delivery of the entire message

- Service point addressing(port address)

Transport Layer header includes service point address which is port address. This layer gets the message to the correct process on the computer which gets each packet to the correct computer.

- Segmentation and reassembly

A message is divided into segments; each segment contains sequence number, which enables this layer in reassembling the message. Message is reassembled correctly upon arrival at the destination and replaces packets which were lost in transmission.

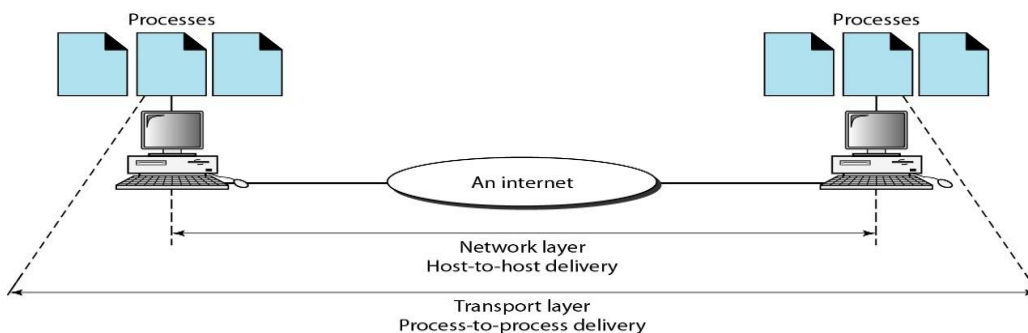
- Connection Control

It provides support or connection oriented and connectionless services.

- Flow Control

- Error Control

Reliable process-to-process delivery of a message



5.Session Layer

- Dialog Control : It allows the communication between two process to take place in either half or full duplex mode.
- Synchronization : Add checkpoints or synchronization points to a stream of data.

6.Presentation layer

- Translation

Before being transmitted, information in the form of characters and numbers should be changed to bit streams. The presentation layer is responsible for

interoperability between encoding methods as different computers use different encoding methods. It translates data between the formats the network requires and the format the computer.

- Encryption

Presentation layer format and encrypts data to be sent across the network. This layer carries out the encryption at the transmitter and decryption at the receiver to enhance security.

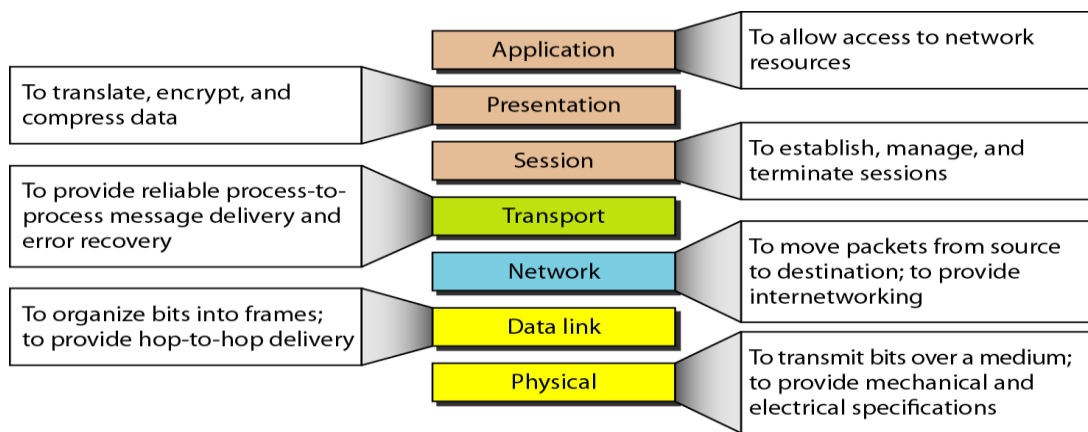
- Compression

This layer carries out data compression to reduce the bandwidth of the data to be transmitted (the primary goal of data compression is to reduce the number of bits which is to be transmitted).

7.Application Layer

Functions:

1. **Mail Services:** This layer provides the basis for E-mail forwarding and storage.
2. **Network Virtual Terminal:** It allows a user to log on to a remote host.
3. **Directory Services:** This layer provides access for global information about various services.
4. **File Transfer, Access and Management (FTAM):** It is a standard mechanism to access files and manages it. Users can access files in a remote computer and manage it. They can also retrieve files from a remote computer.



Internet Protocols

Internet Protocols are a set of rules that governs the communication and exchange of data over the internet. Both the sender and receiver should follow the same protocols in order to communicate the data. In order to understand it better, let's take an example of a language. Any language has its own set of vocabulary and grammar which we need to know if we want to communicate in that language. Similarly, over the internet whenever we access a website or exchange some data with another device then these processes are governed by a set of rules called the internet protocols.

Working of Internet Protocol

The internet and many other data networks work by organizing data into small pieces called packets. Each large data sent between two network devices is divided into smaller packets by the underlying hardware and software. Each network protocol defines the rules for how its data packets must be organized in specific ways according to the protocols the network supports.

Types of Internet Protocol

Internet Protocols are of different types having different uses. They are:

1. TCP/IP (Transmission Control Protocol/ Internet Protocol)
2. SMTP (Simple Mail Transfer Protocol)
3. PPP (Point-to-Point Protocol)
4. FTP (File Transfer Protocol)
5. SFTP (Secure File Transfer Protocol)
6. HTTP (Hyper Text Transfer Protocol)
7. HTTPS (Hyper Text Transfer Protocol Secure)
8. TELNET (Terminal Network)
9. POP3 (Post Office Protocol 3)
10. IPv4
11. IPv6
12. ICMP
13. UDP
14. IMAP
15. SSH
16. Gopher

1. TCP/IP (Transmission Control Protocol/ Internet Protocol)

These are a set of standard rules that allows different types of computers to communicate with each other. The IP protocol ensures that each computer that is connected to the Internet is having a specific serial number called the IP address. TCP specifies how data is exchanged over the internet and how it should be broken into IP packets. It also makes sure that the packets have information about the source of the message data, the destination of the message data, the sequence in

which the message data should be re-assembled, and checks if the message has been sent correctly to the specific destination. The TCP is also known as a connection-oriented protocol.

2. SMTP(Simple Mail Transfer Protocol)

These protocols are important for sending and distributing outgoing emails. This protocol uses the header of the mail to get the email id of the receiver and enters the mail into the queue of outgoing mail. And as soon as it delivers the mail to the receiving email id, it removes the email from the outgoing list. The message or the electronic mail may consider the text, video, image, etc. It helps in setting up some communication server rules.

3. PPP(Point-to-Point Protocol)

It is a communication protocol that is used to create a direct connection between two communicating devices. This protocol defines the rules using which two devices will authenticate with each other and exchange information with each other. For example, A user connects his PC to the server of an Internet Service Provider and also uses PPP. Similarly, for connecting two routers for direct communication it uses PPP.

4. FTP (File Transfer Protocol)

This protocol is used for transferring files from one system to the other. This works on a client-server model. When a

machine requests for file transfer from another machine, the FTO sets up a connection between the two and authenticates each other using their ID and Password. And, the desired file transfer takes place between the machines.

5. SFTP (Secure File Transfer Protocol)

SFTP which is also known as SSH FTP refers to File Transfer Protocol (FTP) over Secure Shell (SSH) as it encrypts both commands and data while in transmission. SFTP acts as an extension to SSH and encrypts files and data then sends them over a secure shell data stream. This protocol is used to remotely connect to other systems while executing commands from the command line.

6. HTTP (Hyper Text Transfer Protocol)

This protocol is used to transfer hypertexts over the internet and it is defined by the www (world wide web) for information transfer. This protocol defines how the information needs to be formatted and transmitted. And, it also defines the various actions the web browsers should take in response to the calls made to access a particular web page. Whenever a user opens their web browser, the user will indirectly use HTTP as this is the protocol that is being used to share text, images, and other multimedia files on the World Wide Web.

7. HTTPS (HyperText Transfer Protocol Secure)

HTTPS is an extension of the Hypertext Transfer Protocol (HTTP). It is used for secure communication over a computer network with the SSL/TLS protocol for encryption and authentication. So, generally, a website has an HTTP protocol but if the website is such that it receives some sensitive information such as credit card details, debit card details, OTP, etc then it requires an SSL certificate installed to make the website more secure. So, before entering any sensitive information on a website, we should check if the link is HTTPS or not. If it is not HTTPS then it may not be secure enough to enter sensitive information.

8. TELNET (Terminal Network)

TELNET is a standard TCP/IP protocol used for virtual terminal service given by ISO. This enables one local machine to connect with another. The computer which is being connected is called a remote computer and which is connecting is called the local computer. TELNET operation lets us display anything being performed on the remote computer in the local computer. This operates on the client/server principle. The local computer uses the telnet client program whereas the remote computer uses the telnet server program.

9. POP3(Post Office Protocol 3)

POP3 stands for Post Office Protocol version 3. It has two Message Access Agents (MAAs) where one is client MAA (Message Access Agent) and another is server MAA(Message Access Agent) for accessing the messages from the mailbox. This protocol helps us to retrieve and manage emails from the mailbox on the receiver mail server to the receiver's computer. This is implied between the receiver and the receiver mail server. It can also be called a one-way client-server protocol.

10. IPv4

The fourth and initially widely used version of the Internet Protocol is called IPv4 (Internet Protocol version 4). It is the most popular version of the Internet Protocol and is in charge of distributing data packets throughout the network. Maximum unique addresses for IPv4 are 4,294,967,296 (2³²), which are possible due to the use of 32-bit addresses. The network address and the host address are the two components of each address. The host address identifies a particular device within the network, whereas the network address identifies the network to which the host belongs. In the “dotted decimal” notation, which is the standard for IPv4 addresses, each octet (8 bits) of the address is represented by its decimal value and separated by a dot (e.g. 192.168.1.1).

11. IPv6

The most recent version of the Internet Protocol, IPv6, was created to address the IPv4 protocol's drawbacks. A maximum of 4.3 billion unique addresses are possible with IPv4's 32-bit addresses. Contrarily, IPv6 uses 128-bit addresses, which enable a significantly greater number of unique addresses. This is significant because IPv4 addresses were running out and there are an increasing number of devices that require internet access. Additionally, IPv6 offers enhanced security features like integrated authentication and encryption as well as better support for mobile devices. IPv6 support has spread among websites and internet service providers, and it is anticipated to gradually displace IPv4 as the main internet protocol.

.

12. ICMP

ICMP (Internet Control Message Protocol) is a network protocol that is used to send error messages and operational information about network conditions. It is an integral part of the Internet Protocol (IP) suite and is used to help diagnose and troubleshoot issues with network connectivity. ICMP messages are typically generated by network devices, such as routers, in response to errors or exceptional conditions encountered in forwarding a datagram. Some examples of ICMP messages include:

- Echo Request and Echo Reply (ping)
- Destination Unreachable
- Time Exceeded

- Redirect

ICMP can also be used by network management tools to test the reachability of a host and measure the round-trip time for packets to travel from the source to the destination and back. It should be noted that ICMP is not a secure protocol, it can be used in some types of network attacks like DDoS amplification.

13. UDP

UDP (User Datagram Protocol) is a connectionless, unreliable transport layer protocol. Unlike TCP, it does not establish a reliable connection between devices before transmitting data, and it does not guarantee that data packets will be received in the order they were sent or that they will be received at all. UDP is faster than TCP because it has less overhead. It doesn't need to establish a connection, so it can send data packets immediately. It also doesn't need to wait for confirmation that the data was received before sending more, so it can transmit data at a higher rate.

14. IMAP

IMAP (Internet Message Access Protocol) is a protocol used for retrieving emails from a mail server. It allows users to access and manage their emails on the server, rather than downloading them to a local device. This means that the user can access their emails from multiple devices and the emails will be synced across all devices. IMAP is more flexible than POP3 (Post Office Protocol version 3) as it allows users

to access and organize their emails on the server, and also allows multiple users to access the same mailbox.

15. SSH

SSH (Secure Shell) is a protocol used for secure remote login and other secure network services. It provides a secure and encrypted way to remotely access and manage servers, network devices, and other computer systems. SSH uses public-key cryptography to authenticate the user and encrypt the data being transmitted, making it much more secure than traditional remote login protocols such as Telnet. SSH also allows for secure file transfers using the SCP (Secure Copy) and SFTP (Secure File Transfer Protocol) protocols. It is widely used in Unix-based operating systems and is also available for Windows. It is commonly used by system administrators, developers, and other technical users to remotely access and manage servers and other network devices.

16. Gopher

Gopher is a type of file retrieval protocol that provides downloadable files with some description for easy management, retrieving, and searching of files. All the files are arranged on a remote computer in a stratified manner. It is an old protocol and it is not much used nowadays.

Application Layer

The application layer in the OSI model is the closest layer to the end user which means that the application layer and end user can interact directly with the software application. The application layer programs are based on client and servers.

The Application layer includes the following functions:

- **Identifying communication partners:** The application layer identifies the availability of communication partners for an application with data to transmit.
- **Determining resource availability:** The application layer determines whether sufficient network resources are available for the requested communication.
- **Synchronizing communication:** All the communications occur between the applications requires cooperation which is managed by an application layer.

Services of Application Layers

- **Network Virtual terminal:** An application layer allows a user to log on to a remote host. To do so, the application creates a software emulation of a terminal at the remote host. The user's computer talks to the software terminal, which in turn, talks to the host. The remote host thinks that it is communicating with one of its own terminals, so it allows the user to log on.
- **File Transfer, Access, and Management (FTAM):** An application allows a user to access files in a remote

computer, to retrieve files from a computer and to manage files in a remote computer. FTAM defines a hierarchical virtual file in terms of file structure, file attributes and the kind of operations performed on the files and their attributes.

- **Addressing:** To obtain communication between client and server, there is a need for addressing. When a client made a request to the server, the request contains the server address and its own address. The server response to the client request, the request contains the destination address, i.e., client address. To achieve this kind of addressing, DNS is used.
- **Mail Services:** An application layer provides Email forwarding and storage.
- **Directory Services:** An application contains a distributed database that provides access for global information about various objects and services.
- **Authentication:** It authenticates the sender or receiver's message or both.

Network Application Architecture

Application architecture is different from the network architecture. The network architecture is fixed and provides a set of services to applications. The application architecture, on the other hand, is designed by the application developer and defines how the application should be structured over the various end systems.

Application architecture is of two types:

1. Client-server architecture: An application program running on the local machine sends a request to another application program is known as a client, and a program that serves a request is known as a server. For example, when a web server receives a request from the client host, it responds to the request to the client host.

Characteristics of Client-server architecture:

- In Client-server architecture, clients do not directly communicate with each other. For example, in a web application, two browsers do not directly communicate with each other.
- A server is fixed, well-known address known as IP address because the server is always on while the client can always contact the server by sending a packet to the sender's IP address.

Disadvantage of Client-server architecture:

It is a single-server based architecture which is incapable of holding all the requests from the clients. For example, a social networking site can become overwhelmed when there is only one server exists.

2. P2P (peer-to-peer) architecture: It has no dedicated server in a data center. The peers are the computers which are not owned by the service provider. Most of the peers reside in the homes, offices, schools, and universities. The peers communicate with each other without passing the information through a dedicated server, this architecture

is known as peer-to-peer architecture. The applications based on P2P architecture includes file sharing and internet telephony.

Features of P2P architecture

- **Self scalability:** In a file sharing system, although each peer generates a workload by requesting the files, each peer also adds a service capacity by distributing the files to the peer.
- **Cost-effective:** It is cost-effective as it does not require significant server infrastructure and server bandwidth.

World Wide Web

The Web is a repository of information in which the documents, called Web pages, are **distributed** all over the world and related documents are **linked** together.

The popularity and growth of the Web can be related to two terms in the above statement: distributed and linked.

Distribution allows the growth of the Web. Each web server in the world can add a new web page to the repository .

Linking allows one web page to refer to another web page stored in another server somewhere else in the world. The linking of web pages was achieved using a concept called **hypertext**.

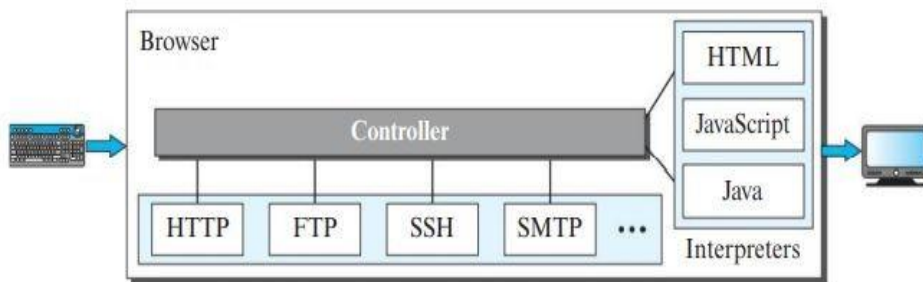
Architecture:

The WWW is a distributed client-server service, in which a client using a browser can access a service using a server.

However, the service provided is distributed over many locations called sites. Each site holds one or more documents, referred to as web pages.

• Web clients(Browsers)

A variety of vendors offer commercial browsers that interpret and display a web page, and all of them use nearly the same architecture. Each browser usually consists of three parts: a controller, client protocols, and interpreters.



- The controller receives input from the keyboard or the mouse and uses the client programs to access the document.
- After the document has been accessed, the controller uses one of the interpreters to display the document on the screen.
- The client protocol can be one of the protocols, such as HTTP or FTP. The interpreter can be HTML, Java, or JavaScript, depending on the type of document.
- Some commercial browsers include Internet

Explorer, Netscape Navigator, and Firefox.

Web Server

The web page is stored at the server. Each time a request arrives, the corresponding document is sent to the client.

Uniform Resource Locator (URL)

A web page, as a file, needs to have a unique identifier to distinguish it from other web pages. To define a web page, we need three identifiers: host, port, and path.

- Host. The host identifier can be the IP address of the server or the unique name given to the server. IP addresses can be defined in dotted decimal notations (such as 64.23.56.17)
- Port. The port, a 16-bit integer, is normally predefined for the client-server application.
- Path. The path identifies the location and the name of the file
- To combine these three pieces together along with protocol, the uniform resource locator (URL) has been designed

protocol://host:port/path

`http://www.example.com/index.html` , which indicates a protocol (`http`), a hostname (`www.example.com`), and a file name (`index.html`).

Web Documents

The documents in the WWW can be grouped into three broad categories: static, dynamic and active .

Static Documents

- Static documents are fixed-content documents that are created and stored in a server.
- The client can get a copy of the document only.
- In other words, the contents of the file are determined when the file is created, not when it is used.

Dynamic Documents

- A dynamic document is created by a web server whenever a browser requests the document.
- When a request arrives, the web server runs an application program or a script that creates the dynamic document.
- The server returns the result of the program or script as a response to the browser that requested the document.
- Because a fresh document is created for each request, the contents of a dynamic document may vary from one request to another.
- A very simple example of a dynamic document is the retrieval of the time and date from a server

Active Documents

- For many applications, it is need a program or a script to be run at the client site. These are called active documents.
- For example, suppose we want to run a program that creates animated graphics on the screen or a program that interacts with the user.
- The program definitely needs to be run at the client site where the animation or interaction takes place. When a browser requests an active document, the server sends a copy of the document or a script. The document is then run at the client (browser) site.

HTTP(Hyper Text Transfer Protocol)

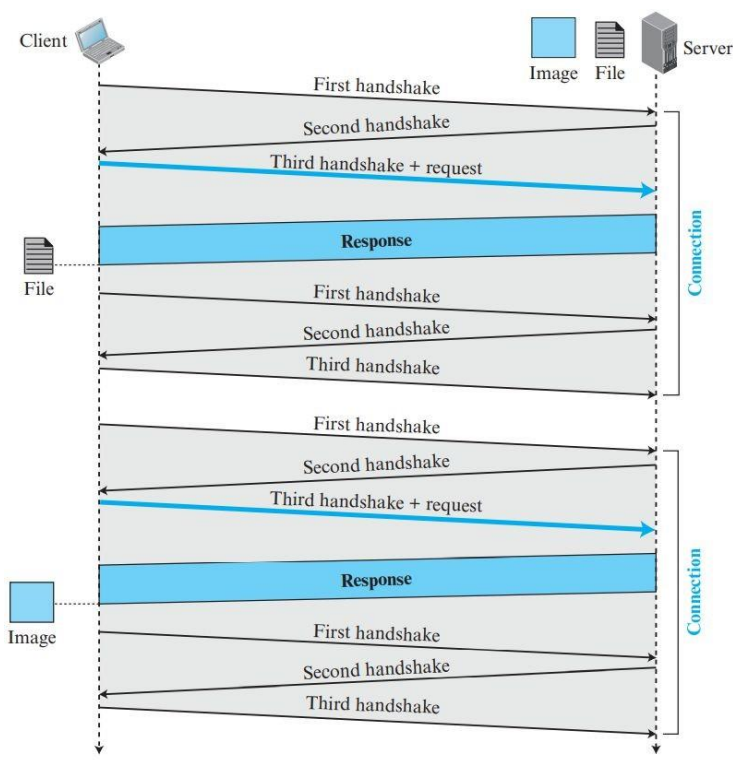
- The HyperText Transfer Protocol (HTTP) is a protocol that is used to define how the client-server programs can retrieve web pages from the Web.
- An HTTP client sends a request; an HTTP server returns a response. The server uses the port number 80; the client uses a temporary port number.
- Before any transaction between the client and the server can take place, connection needs to be established between them.
- After the transaction, the connection should be terminated
- The client and server, however, do not need to worry about errors in messages exchanged or loss of any message, because HTTP relay on the TCP.

If the web pages objects to be retrieved are located on different servers, we don't have any choice that to create a new TCP connection for retrieving each objects. However, if some of the objects are located on the same server, we have two choices: to retrieve each object using a new TCP connection or to make a TCP connection & retrieve them all. The first method is referred to as non-persistent connection and second as persistent connection.

- Non-persistent Connections

In a non-persistent connection, one TCP connection is made for each request/response. The following lists the steps in this strategy:

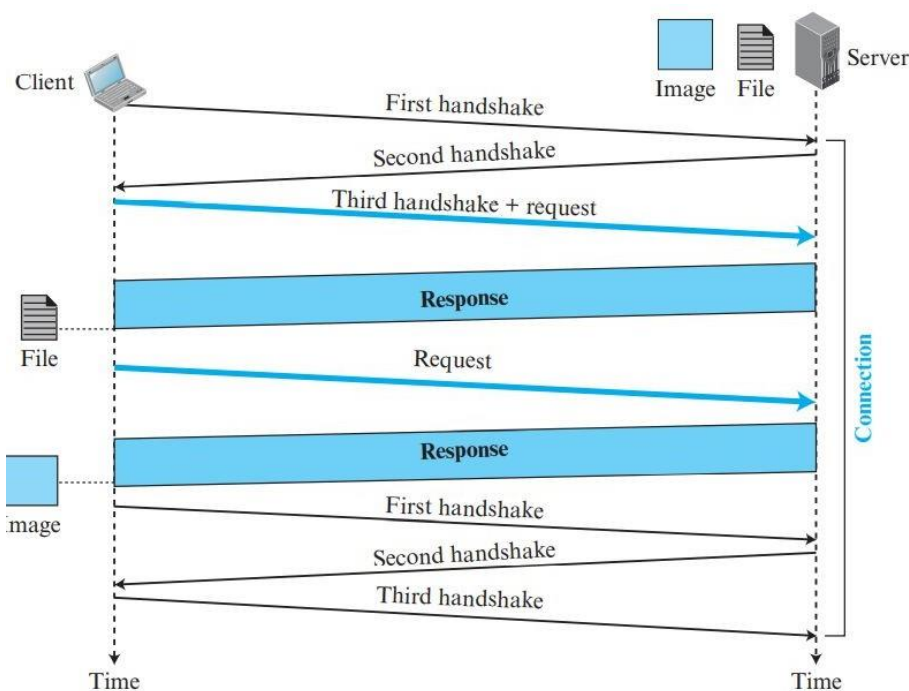
1. The client opens a TCP connection and sends a request.
2. The server sends the response and closes the connection.
3. The client reads the data until it encounters an end-of-file marker; it then closes the connection.



- Persistent Connections

HTTP version 1.1 specifies a persistent connection by default. In a persistent connection, the server leaves the connection open for more requests after sending a response.

The server can close the connection at the request of a client or if a time-out has been reached.

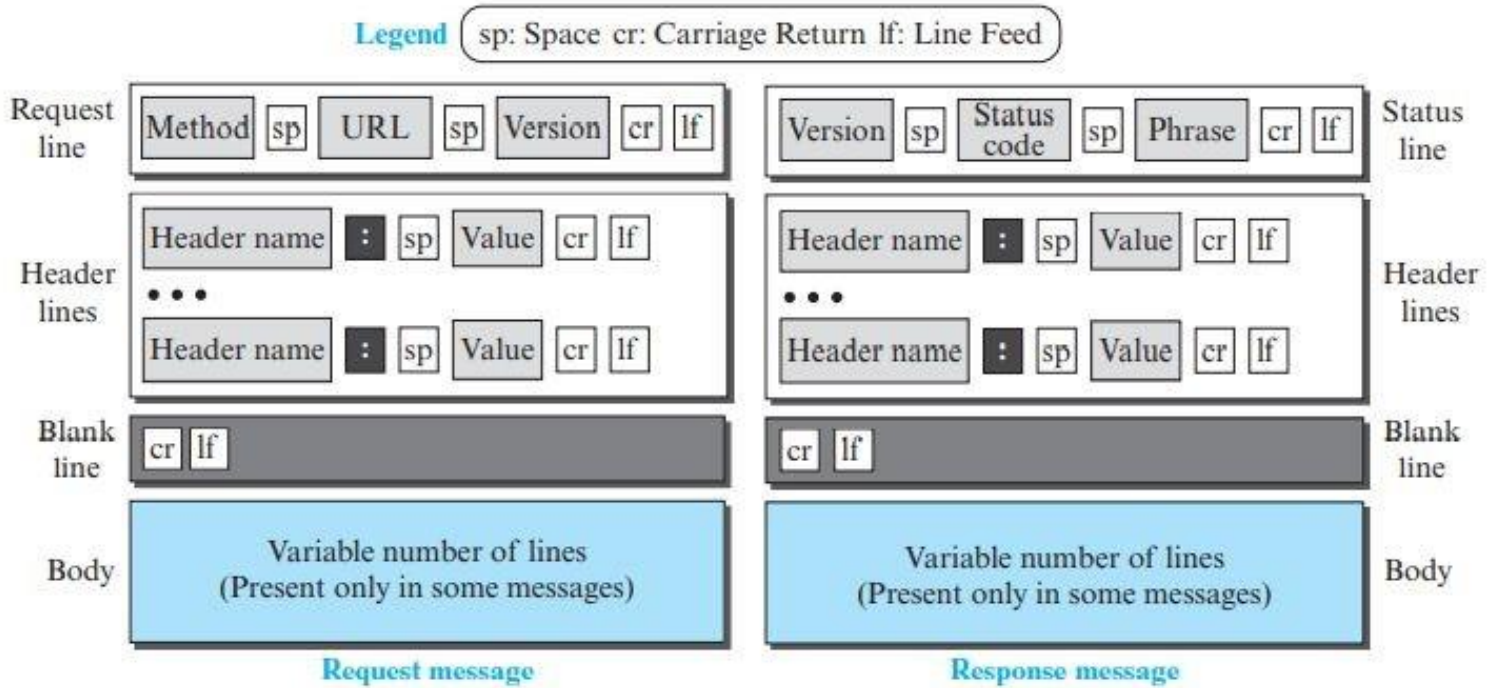


- Message Formats

The HTTP protocol defines the format of the request and response messages.

Each message is made of four sections. The first section in the request message is called the request line; the first section in the response message is called the status line.

The other three sections have the same names in the request and response messages.



Request Message

First line in a request message is called a request line. There are three fields in this line separated by one space and terminated by two characters a carriage return (CR) and a line feed (LF) to specify end of line. The fields are called method, URL, and version.

Method:

<i>Method</i>	<i>Action</i>
GET	Requests a document from the server
HEAD	Requests information about a document but not the document itself
PUT	Sends a document from the client to the server
POST	Sends some information from the client to the server
TRACE	Echoes the incoming request
DELETE	Removes the web page
CONNECT	Reserved
OPTIONS	Inquires about available options

The second field, URL, defines the address and name of the corresponding web page.

The third field, version, gives the version of the protocol; the most current version of HTTP is 1.1

- After the request line, zero or more request header lines. Each header line sends additional information from the client to the server.
- The body can be present in a request message. Usually, it contains the comment to be sent.

Response message

- A response message consists of a status line, header lines, a blank line, and sometimes a body.
- The first line in a response message is called the status line. There are three fields in this line
- The status code field defines the status of the request.
- After the status line, have zero or more response header lines. Each header line sends additional information from the server to the client.
- The body contains the document to be sent from the server to the client.



HTTP Response Status Codes

Status Code	Action	Description
200	OK	Successfully retrieved resource
201	Created	A new resource was created
204	No Content	Request has nothing to return
301 / 302	Moved	Moved to another location (redirect)
400	Bad Request	Invalid request / syntax error
401 / 403	Unauthorized	Authentication failed / Access denied
404	Not Found	Invalid resource was requested
409	Conflict	Conflict was detected, e.g. duplicated email
500 / 503	Server Error	Internal server error / Service unavailable

- Conditional Request

A client can add a condition in its request.

In this case, the server will send the requested web page if the condition is met or inform the client otherwise.

One of the most common conditions imposed by the client is the time and date the web page is modified.

The client can send the header line If-Modified-Since with the request to tell the server that it needs the page only if it is modified after a certain point in time

- Cookies

Web has functions that need to remember some information about the clients; some are listed below:

Websites are being used as electronic stores that allow users to browse through the store, select wanted items, put them in an electronic cart.

Some websites need to allow access to registered clients only. Some websites are used as portals: the user selects the web pages he wants to see. Some websites are just advertising agency.

For these purposes, the cookie mechanism was devised.

Web Caching: Proxy Server

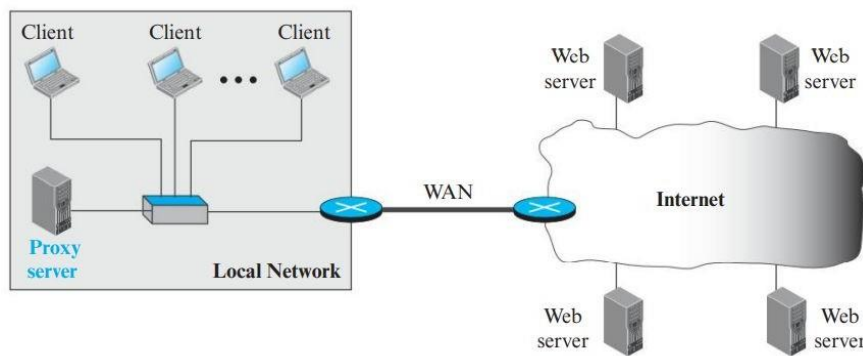
HTTP supports proxy servers.

A proxy server is a computer that keeps copies of responses to recent requests.

The HTTP client sends a request to the proxy server. The proxy server checks its cache.

If the response is not stored in the cache, the proxy server sends the request to the corresponding server.

Incoming responses are sent to the proxy server and stored for future requests from other clients.



- HTTP Security

HTTP does not provide security.

HTTP can be run over the Secure Socket Layer (SSL). In this case, HTTP is referred to as HTTPS. HTTPS provides confidentiality, client and server authentication, and data integrity.

FTP (File Transfer Protocol)

FTP is the standard protocol provided by TCP/IP for copying a file from one host to another.

Although transferring files from one system to another seems simple and straightforward, some problems must be dealt with first.

For example, two systems may use different file name conventions.

Two systems may have different ways to represent data. Two systems may have different directory structures.

All of these problems have been solved by FTP in a very simple and elegant approach.

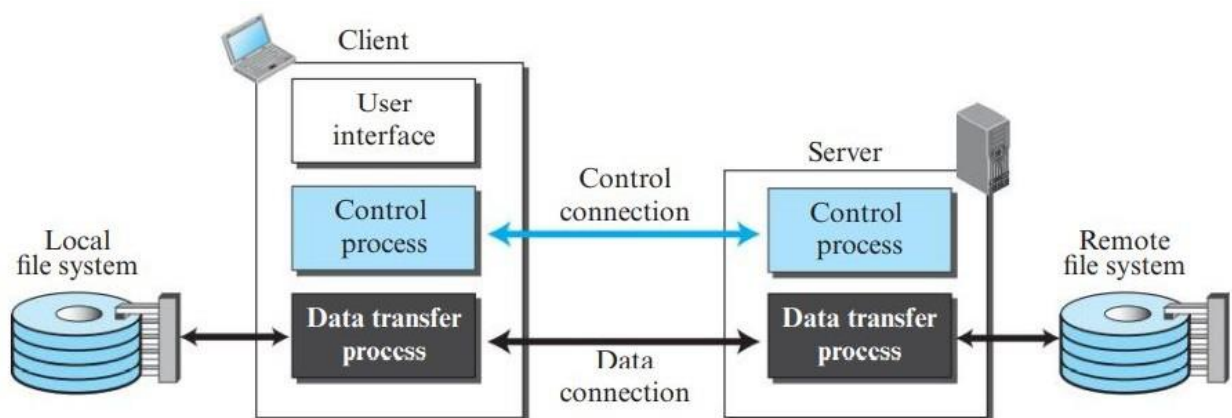


Fig: Basic model of FTP

- The client has three components: user interface, client control process, and the client data transfer process. The server has two components: the server control process and the server data transfer process. The control

connection is made between the control processes. The data connection is made between the data transfer processes.

- Lifetimes of Two Connections

The two connections in FTP have different lifetimes. The control connection remains connected during the entire interactive FTP session. The data connection is opened and then closed for each file transfer activity.

Control Connection:

During this control connection, commands are sent from the client to the server and responses are sent from the server to the client.

Data Connection :

The data connection uses the well-known port 20 at the server site. However, the creation of a data connection is different from the control connection.

The following shows the steps:

1. The client issues a passive open using an ephemeral port. This must be done by the client because it is the client that issues the commands for transferring files.
2. The client sends this port number to the server using the PORT command.
3. The server receives the port number and issues an active open using the well known port 20 and the received ephemeral port number

Data Structure

FTP can transfer a file across the data connection using one of the following interpretations of the structure of the data:

file structure, record structure, or page structure.

The file structure format has no structure. It is a continuous stream of bytes.

In the record structure, the file is divided into records.

In the page structure, the file is divided into pages, with each page having a page number and a page header. The pages can be stored and accessed randomly or sequentially.

Transmission Mode

FTP can transfer a file across the data connection using one of the following three transmission modes: stream mode, block mode, or compressed mode.

The stream mode is the default mode; data are delivered from FTP to TCP as a continuous stream of bytes. In the block mode, data can be delivered from FTP to TCP in blocks.

File Transfer

File transfer occurs over the data connection under the control of the commands sent over the control connection.

Security for FTP

The FTP protocol was designed when security was not a big issue.

Although FTP requires a password, the password is sent in plaintext (unencrypted), which means it can be intercepted and used by an attacker.

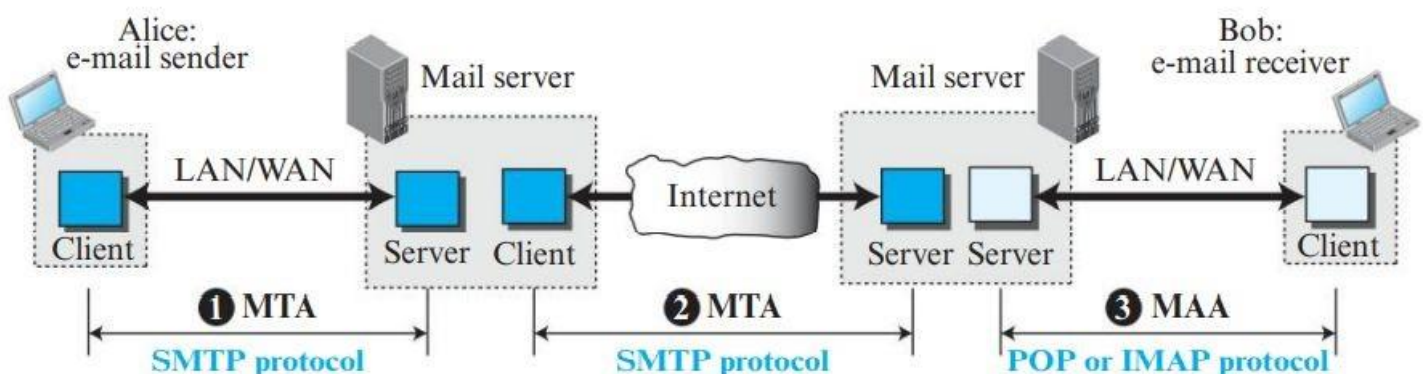
The data transfer connection also transfers data in plaintext, which is insecure.

To be secure, one can add a Secure Socket Layer between the FTP application layer and the TCP layer. In this case FTP is called SSL-FTP

SMTP (Simple Mail Server Protocol)

- Electronic mail (or e-mail) allows users to exchange messages.
- e-mail is one of those applications that needs three uses of client-server paradigms to accomplish its task.
- first and the second as Message Transfer Agents (MTAs), the third as Message Access Agent (MAA).

Protocols used in electronic mail



- The formal protocol that defines the MTA client and server in the Internet is called Simple Mail Transfer Protocol (SMTP).
- SMTP is used two times, between the sender and the sender's mail server and between the two mail servers.
- Commands and Responses: SMTP uses commands and responses to transfer messages between an MTA client and an MTA server. The command is from an MTA client to an MTA server; the response is from an MTA server to the MTA client.
- Commands are sent from the client to the server.
- Responses are sent from the server to the client. A response is a three digit code that may be followed by additional textual information.

Mail Transfer Phases

The process of transferring a mail message occurs in three phases: connection establishment, mail transfer, and connection termination.

1.Connection Establishment

After a client has made a TCP connection to the well known port 25, the SMTP server starts the connection phase. This phase involves the following three steps:

1. The server sends code 220 (service ready) to tell the client that it is ready to receive mail. If the server is not ready, it sends code 421 (service not available).
2. The client sends the HELO message to identify itself, using its domain name address. This step is necessary to inform the server of the domain name of the client.
3. The server responds with code 250 (request command completed) or some other code depending on the situation.

2.Message Transfer

After connection has been established between the SMTP client and server, a single message between a sender and one or more recipients can be exchanged. This phase involves eight steps. Steps 3 and 4 are repeated if there is more than one recipient.

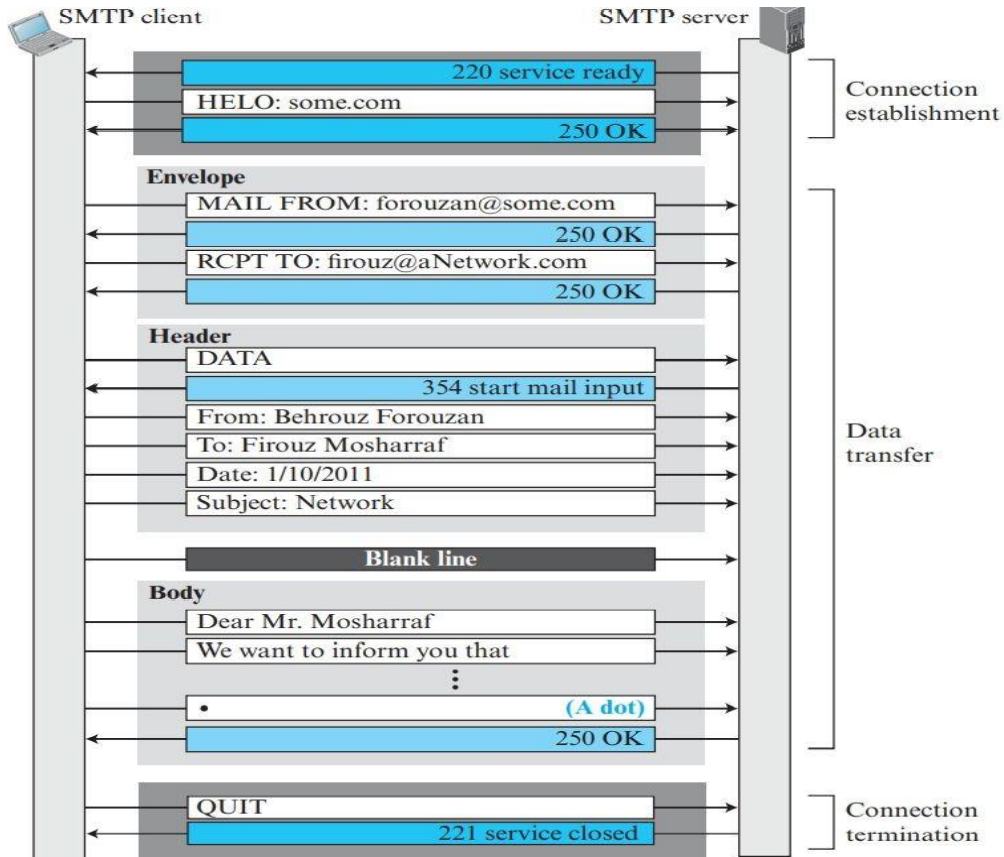
1. The client sends the MAIL FROM message to introduce the sender of the message. It includes the mail address of the sender (mailbox and the domain name). This step is needed to give the server the return mail address for returning errors and reporting messages.
2. The server responds with code 250 or some other appropriate code.

3. The client sends the RCPT TO (recipient) message, which includes the mail address of the recipient.
4. The server responds with code 250 or some other appropriate code.
5. The client sends the DATA message to initialize the message transfer.
6. The server responds with code 354 (start mail input) or some other appropriate message.
7. The client sends the contents of the message in consecutive lines. Each line is terminated by a two-character end-of-line token (carriage return and line feed). The message is terminated by a line containing just one period.
8. The server responds with code 250 (OK) or some other appropriate code.

3.Connection Termination

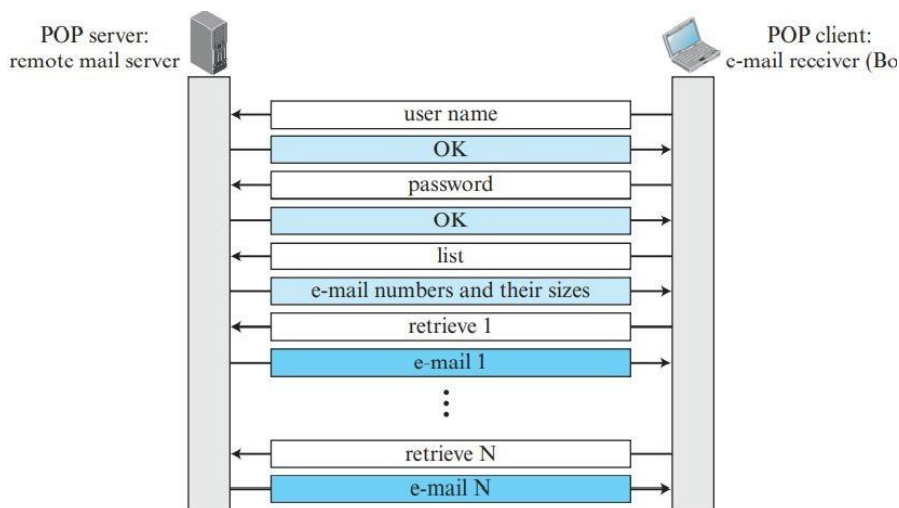
After the message is transferred successfully, the client terminates the connection. This phase involves two steps.

1. The client sends the QUIT command.
2. The server responds with code 221 or some other appropriate code.



POP3(Post Office Protocol, version 3)

- Post Office Protocol, version 3 (POP3) is simple but limited in functionality. The client POP3 software is installed on the recipient computer; the server POP3 software is installed on the mail server.
- Mail access starts with the client when the user needs to download its e-mail from the mailbox on the mail server.
- The client opens a connection to the server on TCP port 110. It then sends its user name and password to access the mailbox. The user can then list and retrieve the mail messages, one by one.



- POP3 has two modes: the delete mode and the keep mode. In the delete mode, the mail is deleted from the mailbox after each retrieval. In the keep mode, the mail remains in the mailbox after retrieval.

DNS(Domain Name System)

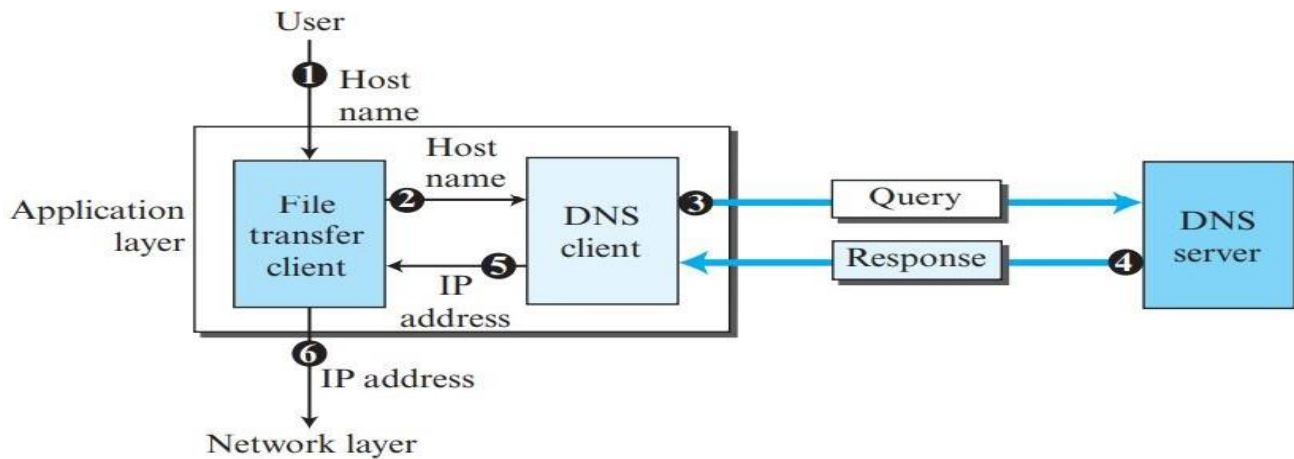
- To identify an entity, TCP/IP protocols use the IP address, which uniquely identifies the connection of a host to the Internet. However, people prefer to use names instead of numeric addresses.
- Since the Internet is so huge today, a central directory system cannot hold all the mapping. In addition, if the central computer fails, the whole communication network will collapse. A better solution is to distribute the information among many computers in the world. In this method, the host that needs mapping can contact the closest computer holding the needed information. This method is used by the Domain Name System (DNS).

The following six steps map the host name to an IP address:

1. The user passes the host name to the file transfer client.
2. The file transfer client passes the host name to the DNS client.
3. Each computer, after being booted, knows the address of one DNS server. The DNS client sends a message to a DNS server with a query that gives the file transfer server name using the known IP address of the DNS server.
4. The DNS server responds with the IP address of the desired file transfer server.

5. The DNS client passes the IP address to the file transfer server.

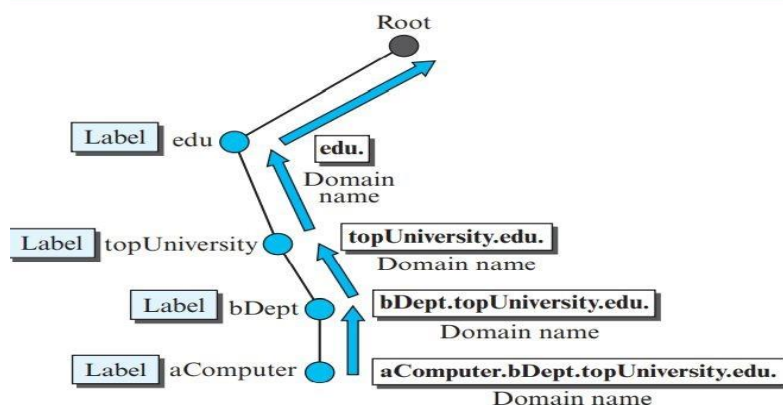
6. The file transfer client now uses the received IP address to access the file transfer server.



A name space that maps each address to a unique name can be organized in two ways: flat or hierarchical.

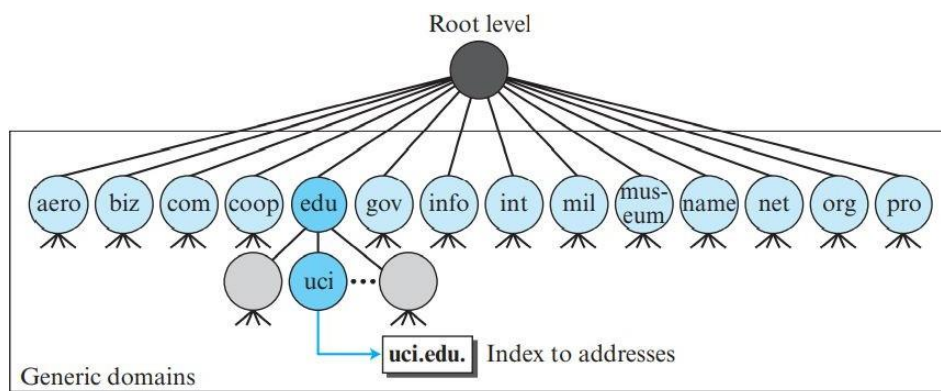
- In a flat name space, a name is assigned to an address. A name in this space is a sequence of characters without structure. The names may or may not have a common section.
- In a hierarchical name space, each name is made of several parts. The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on.

Domain names and labels



DNS in the Internet DNS is a protocol that can be used in different platforms.

- In the Internet, the domain name space (tree) was originally divided into different sections: generic domains, country domains.
- Generic Domains :The generic domains define registered hosts according to their generic behavior.



- Country Domains :The country domains section uses two-character country abbreviations (e.g., us for United States).

Eg:-uci.ca.us.

Resolution

Mapping a name to an address is called name-address resolution.

DNS is designed as a client-server application. A host that needs to map an address to a name or a name to an address calls a DNS client called a resolver.

1. Recursive Resolution

An application program running on a host named some.anet.com needs to find the IP address of another host named engineering.mcgraw-hill.com to send a message to. The source host is connected to the Anet ISP; the destination host is connected to the McGraw-Hill network.

1.The application program on the source host calls the DNS resolver (client) to find the IP address of the destination host.

2.The resolver, which does not know this address, sends the query to the local DNS server (for example, dns.anet.com) running at the Anet ISP site (event 1).

3.Assume that this server does not know the IP address of the destination host either.

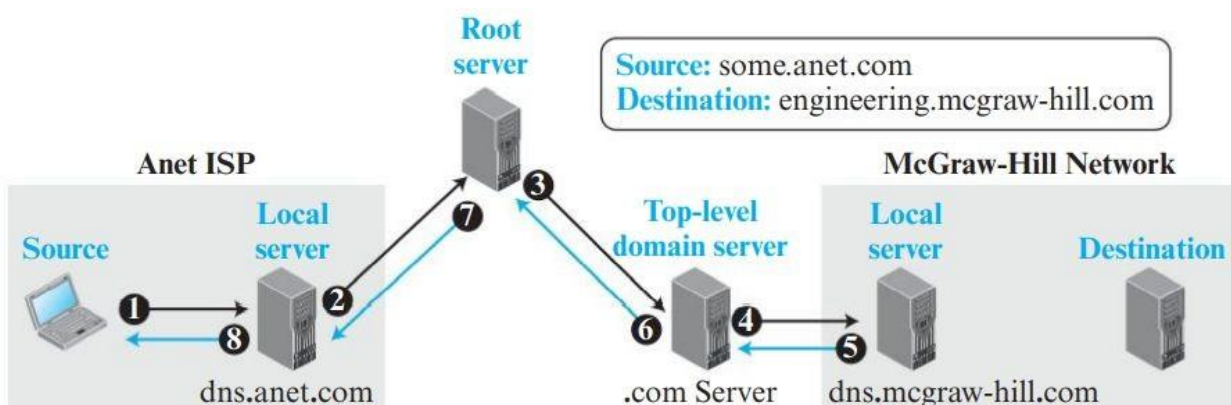
4.It sends the query to a root DNS server, whose IP address is supposed to be known to this local DNS server (event 2).

5. Root servers do not normally keep the mapping between names

and IP addresses, but a root server should at least know about one server at each top level domain (in this case, a server responsible for com domain).

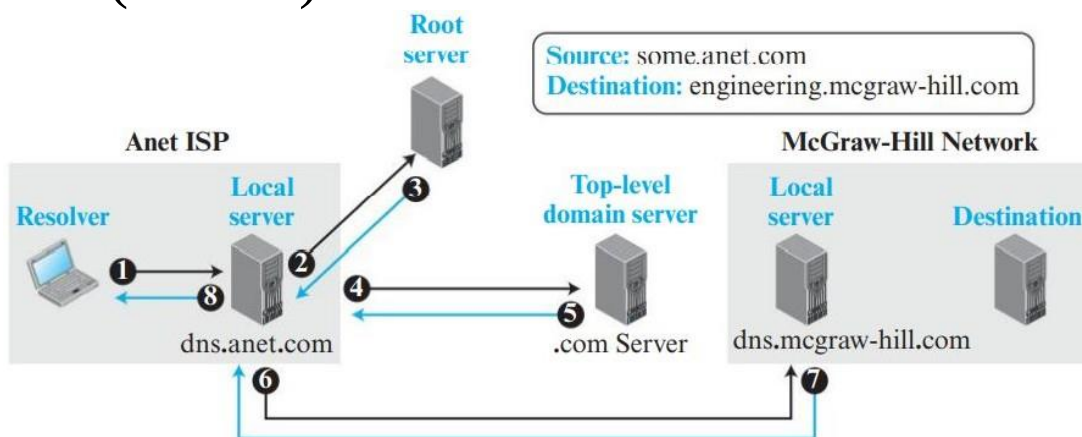
6. The query is sent to this top-level-domain server (event 3). Assume that this server does not know the name-address mapping of this specific destination, but it knows the IP address of the local DNS server in the McGraw-Hill company (for example, dns.mcgraw-hill.com).

7. The query is sent to this server (event 4), which knows the IP address of the destination host. The IP address is now sent back to the top-level DNS server (event 5), then back to the root server (event 6), then back to the ISP DNS server, which may cache it for the future queries (event 7), and finally back to the source host (event 8).



2. Iterative resolution

- Normally the iterative resolution takes place between two local servers; the original resolver gets the final answer from the local server.
- Note that the messages shown by events 2, 4, and 6 contain the same query. However, the message shown by event 3 contains the IP address of the top-level domain server, the message shown by event 5 contains the IP address of the McGraw-Hill local DNS server, and the message shown by event 7 contains the IP address of the destination.
- When the Anet local DNS server receives the IP address of the destination, it sends it to the resolver (event 8).



Caching

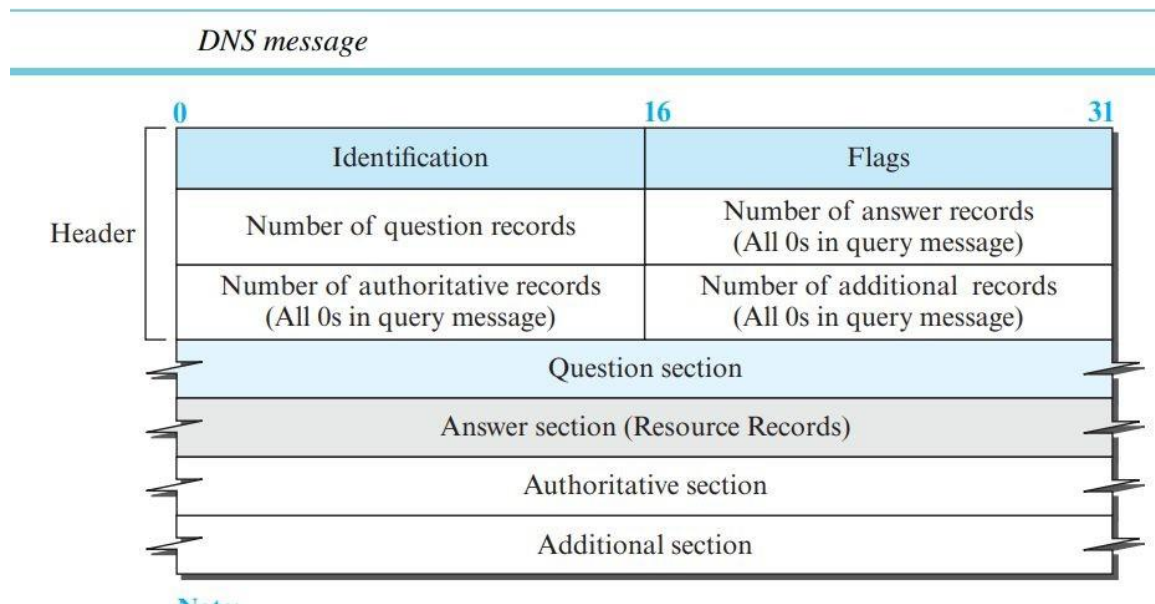
Each time a server receives a query for a name that is not in its domain, it needs to search its database for a server IP address.

Reduction of this search time would increase efficiency. DNS handles this with a mechanism called caching.

When a server asks for a mapping from another server and receives the response, it stores this information in its cache memory before sending it to the client. If the same or another client asks for the same mapping, it can check its cache memory and resolve the problem.

DNS Messages

To retrieve information about hosts, DNS uses two types of messages: query and response.



- The identification field is used by the client to match the response with the query.
 - The flag field defines whether the message is a query or response. It also includes status of error.
 - The next four fields in the header define the number of each record type in the message.
 - The question section, which is included in the query and repeated in the response message, consists of one or more question records.
 - The answer section consist of one or more resource records. It is present only in response messages.
 - The authoritative section gives information (domain name) about one or more authoritative servers for the query.
 - The additional information section provides additional information that may help the resolver.
-
- In DNS, when there is a change, such as adding a new host, removing a host, or changing an IP address, the change must be made to the DNS master file. These types of changes involve a lot of manual updating. The size of today's Internet does not allow for this kind of manual operation. The DNS master file must be updated dynamically. The Dynamic Domain Name System (DDNS) therefore was devised to respond to this need

Peer-peer file sharing networks

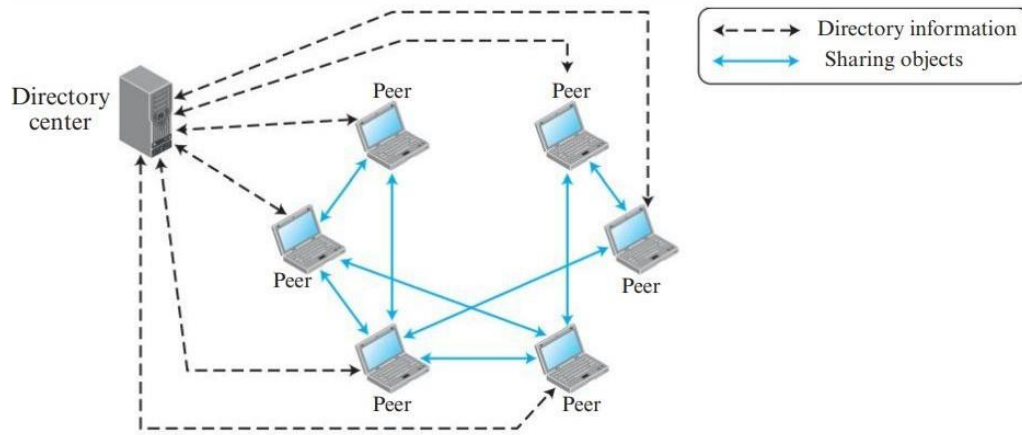
- Internet users that are ready to share their resources become peers and form a network.
- When a peer in the network has a file (for example, an audio or video file) to share, it makes it available to the rest of the peers.
- An interested peer can connect itself to the computer where the file is stored and download it.
- After a peer downloads a file, it can make it available for other peers to download.
- As more peers join and download file, more copies of the file become available to the group.
- P2P networks into two categories: centralized and decentralized.

Centralized Networks

In a centralized P2P network, the directory system uses the client-server paradigm, but the storing and downloading of the files are done using the peer-to-peer paradigm.

For this reason, a centralized P2P network is sometimes referred to as a hybrid P2P network.

Napster was an example of a centralized P2P. In this type of network, a peer first registers itself with a central server. The peer then provides its IP address and a list of files it has to share.



- A peer, looking for a particular file, sends a query to a central server.
- The server searches its directory and responds with the IP addresses of nodes that have a copy of the file.
- The peer contacts one of the nodes and downloads the file. The directory is constantly updated as nodes join or leave the peer.
- Centralized networks make the maintenance of the directory simple but have several drawbacks.
- Accessing the directory can generate huge traffic and slow down the system. The central servers are vulnerable to attack, and if all of them fail, the whole system goes down.

Decentralized Network

- A decentralized P2P network does not depend on a centralized directory system. In this model, peers arrange themselves into an overlay network, which

is a logical network made on top of the physical network. Depending on how the nodes in the overlay network are linked, a decentralized P2P network is classified as either unstructured or structured.

1.Unstructured Networks

In an unstructured P2P network, the nodes are linked randomly. A search in an unstructured P2P is not very efficient because a query to find a file must be flooded through the network, which produces significant traffic and still the query may not be resolved. Two examples of this type of network are Gnutella and Freenet.

2.Structured Networks

A structured network uses a predefined set of rules to link nodes so that a query can be effectively and efficiently resolved.

The most common technique used for this purpose is the Distributed Hash Table (DHT). To avoid the flooding overhead ,DHT-based networks allow each peer to have a partial knowledge about the whole network. This knowledge can be used to route the queries about the data items to the responsible nodes using effective and scalable procedures.

